

BIJAY DHUNGANA

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🎓 [ResearchGate](https://www.researchgate.net/profile/Bijay-Dhungana)

Education

Northern Kentucky University

Bachelor of Science in Computer Science CGPA :3.74/4.00

May 2026

Highland Heights, Kentucky

Experience

Teaching Assistant

Jan 2025 –May 2025

College of Informatics

Highland Heights , Kentucky

- Conducted 3–4 hours of weekly virtual office hours, supporting **60+ students** by guiding them through course concepts and assignments, and addressing individual challenges, leading to a **30%** boost in student engagement.
- Graded weekly class activities, labs, and assignments for over 60 students, ensuring fair, consistent, and timely evaluation while providing constructive feedback that boosted student performance and assignment completion rates by **25%**.
- Collaborated with the course instructor to prepare materials, monitor student progress, and coordinate class activities, while delivering guest lectures streamlining course operations and boosting efficiency by **35%**.

Research Assistant

Oct 2024 –Present

NKU College of Informatics

Highland Heights, Kentucky

- Co-authored a peer-reviewed paper titled *Evaluating Output Novelty in Iterative Prompting in Educational Content Generation*, accepted at ICETC 2025 — proposing a novelty metric that **automated 90.52%** of classification and reduced review time.
- Designed and analyzed experiments using five LLMs (ChatGPT 4o, Claude 3.7, Gemini 2.5 Flash, DeepSeek v3, and r1), benchmarking model efficiency and improving course outline generation accuracy by **35%**.
- Implemented and analyzed lexical and semantic similarity methods **Jaccard**, **CosinE**, **BERT embeddings**, and **Jaro-Winkler** distance to evaluate novelty in LLM-generated course outlines, improving classification accuracy by **35%**.

Projects

AI powered Expense Tracker | Python, React, Firebase

Jan 2025

- Developed a personal finance web application with **Firebase** Authentication for secure user login, role-based access control, and interactive dashboards for managing expenses.
- Integrated **OpenAI API** to automatically categorize expenses and generate personalized budgeting recommendations.
- Designed interactive data visualizations using **Recharts/D3.js** for real-time expense tracking and insights.
- Implemented responsive mobile-first design and deployed through GitHub with **CI/CD** pipeline for seamless updates.

Wanderer - Travel Companion Matching App | React, Typescript, Firebase

June 2025

- Developed an AI-driven travel companion platform connecting travelers through matching algorithms, automated trip planning with a **multi-agent AI** system, and real-time messaging for coordination.
- Built scalable full-stack architecture with a React/TypeScript frontend, Node.js/Express backend, Firebase cloud infrastructure, and **RESTful** APIs supporting user profiles, messaging, and map integration.
- Optimized user experience with **AI-powered** travel planning, real-time status updates/notifications, and responsive design, improving responsiveness and user engagement.

Technical Skills

Languages and data bases:Python, Java, C++, JavaScript, SQL, HTML/CSS, PostgreSQL, MySQL

Frameworks Libraries: Spring Boot, Django, Flask, React.js,Node.js, Pandas, NumPy

Cloud Deployment:AWS, Azure, Firebase

Leadership and Honors

Facility Manager

Dec 2023 –Present

NKU Campus Recreation Center

Highland Heights, Kentucky

NKU Ambassador Scholarship

Aug 2022-Present

Northern Kentucky University

Highland Heights, Kentucky

NKU Presidential List

Dec 2024

College of Informatics,NKU

Highland Heights ,Kentucky